Low-Rank Representation Learning With Spectral-to-Spatial Aggregation Transformer for Hyperspectral Image Super-Resolution

Shibo Wang, Zhichang Guo, Qiting Li, Yao Li, and Yuming Xing

***Abstract*—Hyperspectral image (HSI) faces significant challenges in achieving high-resolution (HR) due to the inherent limitations of imaging devices. To tackle this issue, HSI super-resolution (SR) techniques have been introduced to reconstruct HR images from low-resolution (LR) inputs, demonstrating remarkable progress in enhancing spatial fidelity. Notably, the spectral domain of HSIs inherently exhibits low-rank characteristics resulting from strong interband correlations. Inspired by this observation, many approaches have been developed to leverage the low-rank property of HSIs for performance improvement. However, most existing methods rely on linear low-rank formulations, which are often insufficient to capture the complex spectral structures of HSIs. Moreover, approaches relying on singular value decomposition (SVD) or tensor factorization usually require explicit decomposition operations, which may increase optimization complexity. In this article, we propose a nonlinear low-rank representation (LRR) learning strategy combined with a spectral-to-spatial aggregation Transformer for HSI SR tasks. Specifically, the learned nonlinear LRR is embedded into the spatial attention module to exploit both spatial self-similarity and spectral low-rank characteristics, enabling compact and informative representations that preserve fine spatial and spectral details. In parallel, the spectral-to-spatial aggregation Transformer bridges the spectral and spatial domains, enabling more effective integration of spectral priors into spatial feature learning. Furthermore, we incorporate a spectral convolution attention module into our Transformer framework to mitigate the limitation of Transformers in extracting high-frequency information. Experimental results demonstrate that our method effectively reconstructs SR images with sharper edges, finer details, and enhanced spatial–spectral consistency, outperforming existing approaches in both qualitative and quantitative evaluations.**

***Index Terms*—High-frequency information, hyperspectral image (HSI) super-resolution (SR), low-rank representation (LRR), spectral-to-spatial aggregation transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide richer spectral information compared to conventional RGB or multispectral imagery, which are constrained by limited spectral coverage. By capturing tens to hundreds of contiguous spectral bands, HSIs enable more accurate material identification and comprehensive scene analysis, and have been widely applied in various fields, such as target detection [1], remote sensing [2], and geological exploration [3]. However, HSIs suffer from an inherent trade-off imposed by physical and sensor constraints: increasing spectral resolution typically leads to reduced spatial resolution, which can substantially limit their practical applicability. To alleviate this issue, image super-resolution (SR) techniques have attracted increasing attention. In natural image SR, general-purpose architectures have achieved remarkable success, evolving from deep convolutional networks [4] to advanced vision transformers [5], [6], [7]. Moreover, physically inspired designs, such as fluid micelle networks [8] and heat transfer-inspired models [9], have further demonstrated strong reconstruction capability for RGB image SR. Nevertheless, directly extending these general-purpose SR models to HSI SR is nontrivial, because HSI reconstruction requires not only recovering fine spatial details but also preserving spectral fidelity and cross-band consistency. Therefore, specialized HSI SR methods are still essential for enhancing spatial resolution while maintaining faithful spectral information.

Existing HSI SR methods can be broadly divided into fusion-based and single-image approaches. Fusion-based methods enhance the spatial resolution of HSIs by injecting high-frequency details from auxiliary high-resolution (HR) images, such as panchromatic [10], [11], [12] or multispectral images [13], [14], and have been extensively studied [15], [16]. However, they rely on accurately registered auxiliary data, which is often difficult to obtain in practice. Consequently, single-image HSI SR methods [17], [18], [19], [20], [21], [22], [23], which recover an HR HSIs from only a low-resolution (LR) observation, have attracted increasing attention. Single-image HSI SR methods can be further categorized into model-based and deep learning-based approaches. Model-based methods exploit handcrafted priors, such as total variation (TV) [24], nonlocal similarity [25], and low-rank properties [26], but often suffer from high computational cost, limited flexibility, and difficulty in modeling complex spatial–spectral structures. By contrast, deep learning-based methods learn data-driven priors that better capture spatial–spectral correlations and generally achieve higher reconstruction accuracy.

Deep learning-based HSI SR has evolved from convolution neural network (CNN)-based models to Transformer-based architectures, where the latter are appealing for modeling long-

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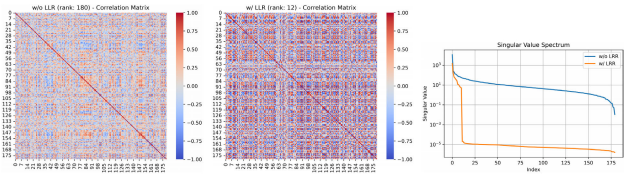
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(a) (b) (c)

Fig. 1. Effect of LRR on feature correlation and singular value spectrum (a) Feature correlation matrix without LRR. (b) Feature correlation matrix with LRR. (c) Singular value spectrum with and without LRR.

range dependencies across spatial and spectral dimensions [23], [27], [28], [29]. Nevertheless, most existing methods still treat spatial and spectral modeling as largely decoupled components [19], [30], which limits the exploitation of intrinsic spectral–spatial interactions in HSIs. Several recent methods attempt to facilitate spectral–spatial information exchange [20], [31], yet they typically do not explicitly exploit the inherent spectral low-rank structure induced by strong interband correlations in HSIs. To exploit this inherent spectral low-rank property, prior studies have introduced explicit low-rank modeling to capture spectral redundancy. For example, Cao et al. [32] proposed a spatial–spectral low-rank representation (LRR) based on robust principal component analysis (PCA), and Zhang et al. [33] employed Tucker tensor factorization to impose global spectral low-rank constraints for HSI SR. These methods typically rely on explicit singular value decomposition (SVD) or tensor decompositions, which may increase optimization complexity and limit their compatibility with lightweight, end-to-end deep networks. More recently, Zhang et al. [34] integrated the learned tensor low-rank priors into an iterative optimization algorithm to obtain an end-to-end architecture. Despite their effectiveness, these low-rank modeling methods are built on linear low-rank formulations and mainly emphasize local feature aggregation, limiting their ability to model complex nonlinear and long-range spectral–spatial dependencies.

In practice, HSI spectra often deviate from a simple global linear structure, owing to spectral heterogeneity caused by material diversity and scene variability, as well as the additional complexity introduced by nonlinear mixing. As a result, a single global linear low-rank model may either require a relatively high rank to fit such heterogeneous spectra or produce unsatisfactory approximations for certain materials and regions. To overcome this limitation, we propose a nonlinear LRR learning strategy that models spectra using learnable bases and nonlinear mixing weights, thereby providing a more flexible compact representation without resorting to explicit SVD or tensor factorization. Specifically, the LRR is implemented as a learnable basis tensor, optimized in an end-to-end manner and incorporated into the spatial attention through a low-rank spatial attention module (LRSAM). When LRSAM is integrated into a spectral-to-spatial aggregation Transformer, the resulting model, termed the low-rank spectral-to-spatial aggregation transformer (LRS2SAT), forms an efficient framework that exploits spectral low-rank correlations while better modeling complex nonlinear spectral–spatial relationships. As shown in Fig. 1, the feature correlation matrix exhibits much stronger interfeature dependencies after applying LRR, as evidenced by the increased density of off-diagonal elements in Fig. 1(b) compared with Fig. 1(a). To quantify this compactness, we compute the effective rank (eRank), defined as the number of singular values greater than 10^{-5}, which decreases sharply from 180 (without LRR) to 12 (with LRR). The singular value spectrum in Fig. 1(c) further shows a steeper decay, indicating that LRR suppresses redundant components and enforces the low-rank property of HSI features, thereby providing informative guidance for the network and leading to better reconstruction quality.

However, Transformer architectures have been shown to favor low-frequency information and are less effective than CNNs at recovering high-frequency details [35]. To mitigate this issue, we introduce a spectral convolution attention module (SCAM), a lightweight spectral convolution branch that injects complementary high-frequency cues for fine-detail reconstruction. We further construct a high-frequency complementary transformer (HFCT) block by coupling LRS2SAT with SCAM in parallel, where the spectral convolution branch complements the Transformer branch with high-frequency information. As shown in Fig. 2, the overall network, which combines multiple HFCT blocks together with convolutional layers and an upsampling module, is termed the LRS2ST. A preliminary version of this work was published in [36]. This manuscript substantially extends the conference version by proposing a novel LRR learning strategy, introducing LRS2SAT and SCAM, and providing expanded experiments and analyses. Our main contributions are summarized as follows.

1) We propose a novel LRR learning strategy that exploits intrinsic nonlinear spectral correlations in HSIs to extract compact low-dimensional features while preserving spectral fidelity, thereby enabling efficient spectral–spatial representations without relying on computationally expensive SVD or explicit tensor decompositions.
2) We introduce LRS2SAT, which employs a spectral-to-spatial attention mechanism to inject spectral guidance into spatial feature learning and enhance representation capability.
3) We propose a local eRank evaluation strategy to assess whether the reconstructed results preserve the low-rank property and maintain consistency with the HR reference. Extensive experiments further demonstrate the effectiveness of the proposed method.

II. Related Works

A. Single HSI SR

Given the superior feature extraction capability of CNNs, numerous CNN-based architectures have been proposed for HSI SR tasks. Li et al. [37] introduced a method that combines spatial constraints with a spectral difference CNN. To capture global spectral correlations, Mei et al. [38]

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proposed a 3-D fully CNN that jointly exploits spatial textures and spectral relationships, but at the cost of heavy computation for high upsampling factors. To reduce this overhead, Li et al. [39] decomposed the 3-D convolutional kernel into separate 1-D and 2-D kernels. Li et al. [40] further introduced a hybrid recursive network combining 2-D convolution with decomposed 3-D convolution, effectively reducing structural redundancy and enhancing spatial–spectral domain learning. Specifically, the grouping strategy was introduced as an alternative to traditional 3-D convolution to efficiently model correlations between adjacent spectral bands. Li et al. [18] developed a grouped deep recursive residual network (GDRRN) by embedding group-wise convolution into recursive residual modules. Additionally, Jiang et al. [41] proposed a progressive upsampling framework utilizing grouped convolution with shared network parameters, thereby improving performance and simplifying feature extraction. To better incorporate HSI characteristics, Liu et al. [17] proposed EUNet, an unfolding network with disentangled spatial–spectral representations. Furthermore, Liu et al. [22] developed SRDNet [22], which combines hybrid convolution, progressive upsampling, and a hyperspectral frequency loss to exploit both spatial–spectral and frequency-domain information. However, these CNN-based approaches mainly capture local features and remain limited in modeling long-range dependencies essential for HSI reconstruction.

Recently, Transformer-based models have been introduced for HSI SR to capture long-range dependencies via self-attention. Chen et al. [42] developed a multiscale deformable Transformer to exploit local and global multiscale spatial–spectral information, while Tang et al. [21] presented a CNN–Transformer unfolding network that integrates spatial–spectral priors guided by HSI degradation characteristics to improve reconstruction quality. To better exploit the spectral information in HSIs, Zhang et al. [23] replaced the conventional attention similarity with the spectral correlation coefficient and further adopted a kernelizable attention formulation, reducing the computational complexity of self-attention from quadratic to linear. Moreover, Chen et al. [19] introduced a cross-scope spatial–spectral Transformer to efficiently capture long-range dependencies, and Xu et al. [20] proposed a spatial–spectral interactive Transformer U-Net with alternating sampling for comprehensive feature extraction, albeit at relatively high computational cost. However, existing Transformer-based methods do not fully exploit intrinsic spatial–spectral correlations and often overlook the inherent spectral low-rank property, which limits reconstruction performance. In this article, we exploit the inherent low-rank property of HSIs to reconstruct the super-resolved HSI.

B. Low-Rank-Based HSI Restoration

Low-rank models have been extensively investigated for HSI restoration. Matrix-based low-rank recovery methods, which represent HSIs as 2-D matrices, have demonstrated remarkable effectiveness in exploiting the intrinsic low-dimensional structures embedded in high-dimensional HSI data. Zhang et al. [43] formulated the HSI restoration problem within a low-rank matrix recovery framework by lexicographically rearranging an HSI patch into a 2-D matrix. In general, HSIs are assumed not only to reside in a low-rank subspace from the spectral perspective but also to exhibit piecewise smoothness in the spatial domain. Based on this prior, He et al. [24] proposed a mixed-noise removal method for HSIs by integrating low-rank matrix factorization with TV regularization. However, representing an HSI as a 2-D matrix inevitably destroys its inherent spatial–spectral structure. To better preserve the spatial–spectral correlations of HSIs, tensor-based low-rank models have also been introduced for HSI restoration. Wang et al. [44] proposed a tensor-based HSI restoration method that combines Tucker decomposition, anisotropic spatial–spectral TV regularization, ℓ_1 norm regularization, and the Frobenius norm to characterize clean HSI structures and mixed noise. Chen et al. [45] and Peng et al. [46] both developed Tucker decomposition-based low-rank tensor models for HSI SR to exploit the spatial–spectral correlation of HSIs. Specifically, Chen et al. [45] incorporated weighted group sparsity regularization as a complementary prior, whereas Peng et al. [46] further exploited nonlocal similarity by modeling grouped nonlocal similar cubes as low-rank 3-D tensors. Wu et al. [47] proposed a low-rank matrix estimation method for HSI SR, where the HSR problem was formulated as a global–local rank-regularized least-squares model by exploiting the low-rank structures of both the whole image and local image regions. However, these methods usually depend on SVD or tensor decomposition, which are computationally expensive. Recently, Zhang et al. [34] proposed a deep neural network for HSI reconstruction by learning the tensor low-rank prior in the feature domain, inspired by the canonical polyadic (CP) decomposition theory, and integrating the learned prior into an iterative optimization framework to build an end-to-end architecture. While these methods have achieved encouraging results, their reliance on linear low-rank formulations and local feature aggregation limits their ability to characterize complex nonlinear relationships and long-range spectral–spatial dependencies.

C. Representation Learning

Representation learning, a subfield of machine learning, has received considerable attention in recent years for its capability to automatically extract meaningful features directly from raw data, thereby eliminating the need for manual feature engineering. Previous works have employed contrastive learning to enhance data representations by comparing the similarities between different samples, enabling models to extract more discriminative and informative features. For instance, Wang et al. [48] introduced DASR, leveraging contrastive learning to propose an unsupervised degradation representation learning framework for blind SR, thereby removing the necessity for explicit degradation estimation. Similarly, Li et al. [49] proposed AirNet, utilizing contrastive learning to address all-in-one image restoration tasks without prior knowledge of corruption types or severity levels. Recently, Yao et al.

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[50] presented a neural degradation representation (NDR) to capture the intrinsic characteristics of diverse image degradations for universal image restoration. Their method adaptively learns extensive degradation features, thus facilitating effective restoration across various degradation types.

Motivated by these works, we propose a LRR learning strategy that adaptively learns and leverages the inherent low-rank properties among spectral bands without incurring additional computational overhead. By effectively capturing the intrinsic low-dimensional structures embedded within high-dimensional spectral data, our approach enhances feature representations, reduces redundancy, and ultimately improves the reconstruction performance in HSI SR tasks.

III. Preliminary

A. Low-Rank Approximation and PCA

Low-rank approximation provides a compact representation for high-dimensional data and serves as a common foundation for HSI restoration. In many low-rank-based methods, an HSI is assumed to lie in a low-dimensional spectral subspace, such that rank minimization can be approximated by low-rank factorization. Let $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ denote an HSI, where H, W, and B represent the spatial height, spatial width, and number of spectral bands, respectively. Under this assumption, $\mathcal{X}$ can be expressed as

$$\mathcal{X} = \mathcal{Z} \times_3 \mathbf{E} \tag{1}$$

where $\mathbf{E} \in \mathbb{R}^{B\times K}$ ($K \ll B$) is the endmember matrix and $\mathcal{Z} \in \mathbb{R}^{H\times W\times K}$ is the abundance tensor.

As a result, each pixel spectrum can be represented as a linear combination of the K columns of $\mathbf{E}$, and therefore lies in a K-dimensional spectral subspace. By stacking all pixel spectra into the mode-3 unfolding matrix $\mathbf{X}_{(3)} \in \mathbb{R}^{HW\times B}$, we obtain a matrix with rank at most K. This indicates that $\mathbf{X}_{(3)}$ admits a LRR. Accordingly, one can model it by a rank-K approximation, leading to the following classical low-rank matrix approximation problem

$$\min_{\operatorname{rank}(\mathbf{X}_K)\le K} \left\|\mathbf{X}_{(3)} - \mathbf{X}_K\right\|_F^2. \tag{2}$$

By the Eckart–Young–Mirsky theorem, the optimal solution is given by the truncated SVD. Let $\mathbf{X}_{(3)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the SVD of $\mathbf{X}_{(3)}$. Then, the best rank-K approximation is

$$\mathbf{X}_K = \mathbf{U}_K\mathbf{\Sigma}_K\mathbf{V}_K^\top \tag{3}$$

where $\mathbf{U}_K$, $\mathbf{\Sigma}_K$, and $\mathbf{V}_K$ contain the top K singular vectors and singular values. Equivalently, this approximation can be written as

$$\mathbf{X}_K = \mathbf{X}_{(3)}\mathbf{V}_K\mathbf{V}_K^\top \tag{4}$$

which admits a projection–reconstruction interpretation: $\mathbf{X}_{(3)}$ is first projected onto the K-dimensional subspace spanned by the leading right singular vectors $\mathbf{V}_K$, and then reconstructed in the original space. This paradigm underlies a wide range of methods for data compression, denoising, and representation learning.

A representative application of low-rank approximation is PCA. Given centered data points $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$, we stack them into a data matrix

$$\mathbf{Y} = \left[\mathbf{x}_1^\top, \mathbf{x}_2^\top, \ldots, \mathbf{x}_n^\top\right]^\top \in \mathbb{R}^{n\times d}. \tag{5}$$

PCA seeks an orthonormal basis $\mathbf{W} \in \mathbb{R}^{d\times \ell}$ with $\ell \le d$ that minimizes the total reconstruction error

$$\min_{\mathbf{W}^\top\mathbf{W}=\mathbf{I}_\ell} \sum_{i=1}^{n} \left\|\mathbf{x}_i - \mathbf{W}\mathbf{W}^\top\mathbf{x}_i\right\|_2^2 = \min_{\mathbf{W}^\top\mathbf{W}=\mathbf{I}_\ell} \left\|\mathbf{Y} - \mathbf{Y}\mathbf{W}\mathbf{W}^\top\right\|_F^2. \tag{6}$$

The optimal $\mathbf{W}$ is given by the top ℓ eigenvectors of the covariance matrix $\mathbf{Y}^\top\mathbf{Y}$, equivalently, the leading right singular vectors of $\mathbf{Y}$, yielding a compact representation that preserves the dominant variances. Despite its effectiveness, PCA is a linear model that uses a fixed set of bases and applies the same projection $\mathbf{Y}\mathbf{W}\mathbf{W}^\top$ to all samples once learned. This global linear constraint can be restrictive for real-world visual data, where spatially varying patterns may require more adaptive and nonlinear representations. Moreover, PCA usually involves an explicit decomposition step, such as SVD or eigendecomposition, which increases optimization complexity and makes it less straightforward to integrate into lightweight end-to-end deep networks.

To overcome these limitations, we adopt a more flexible formulation of the form $f(\mathbf{Y}\mathbf{W})\mathbf{W}^\top$, where $f(\cdot)$ is a nonlinear reweighting function. This formulation replaces the fixed linear projection in PCA-style low-rank modeling with a nonlinear and input-adaptive representation mapping. Moreover, unlike kernel PCA, it does not require explicit kernel matrices or eigendecomposition, and can be implemented as a lightweight, end-to-end trainable block using standard linear layers and pointwise operations. Concretely, we instantiate this idea as an LRR module that generalizes PCA by learning a basis matrix R and its input-dependent combinations

$$\mathbf{F}_{\mathrm{LR}} = \phi(\mathbf{F}\mathbf{R})\,\mathbf{R}^\top \tag{7}$$

where ϕ denotes the softmax operation applied along the basis dimension, $\mathbf{F} \in \mathbb{R}^{C\times M}$ is the input feature matrix (e.g., obtained by unfolding a feature tensor into C spatial locations with M channels), $\mathbf{R} \in \mathbb{R}^{M\times N}$ is a learnable basis matrix, and $\mathbf{F}_{\mathrm{LR}} \in \mathbb{R}^{C\times M}$ is the low-rank output with rank at most N. This formulation can be viewed as a data-driven generalization of PCA: it retains an explicit low-rank factorization form while enabling nonlinear and input-adaptive basis combinations, thereby bridging classical low-rank modeling and end-to-end deep neural architectures.

Moreover, the adoption of softmax is motivated by two considerations. First, from the representation perspective, softmax emphasizes dominant bases while suppressing weak responses, yielding a more concentrated and low-entropy basis distribution. This reduces the effective number of active bases, lowers the effective complexity of each latent code, and makes different spatial locations more likely to share similar basis-assignment patterns, which is beneficial for compact low-rank reconstruction. Second, from a variational perspective [51], softmax is the closed-form solution to an entropy-regularized optimization problem on the simplex. It can therefore be interpreted as a smooth, differentiable, and robust approximation

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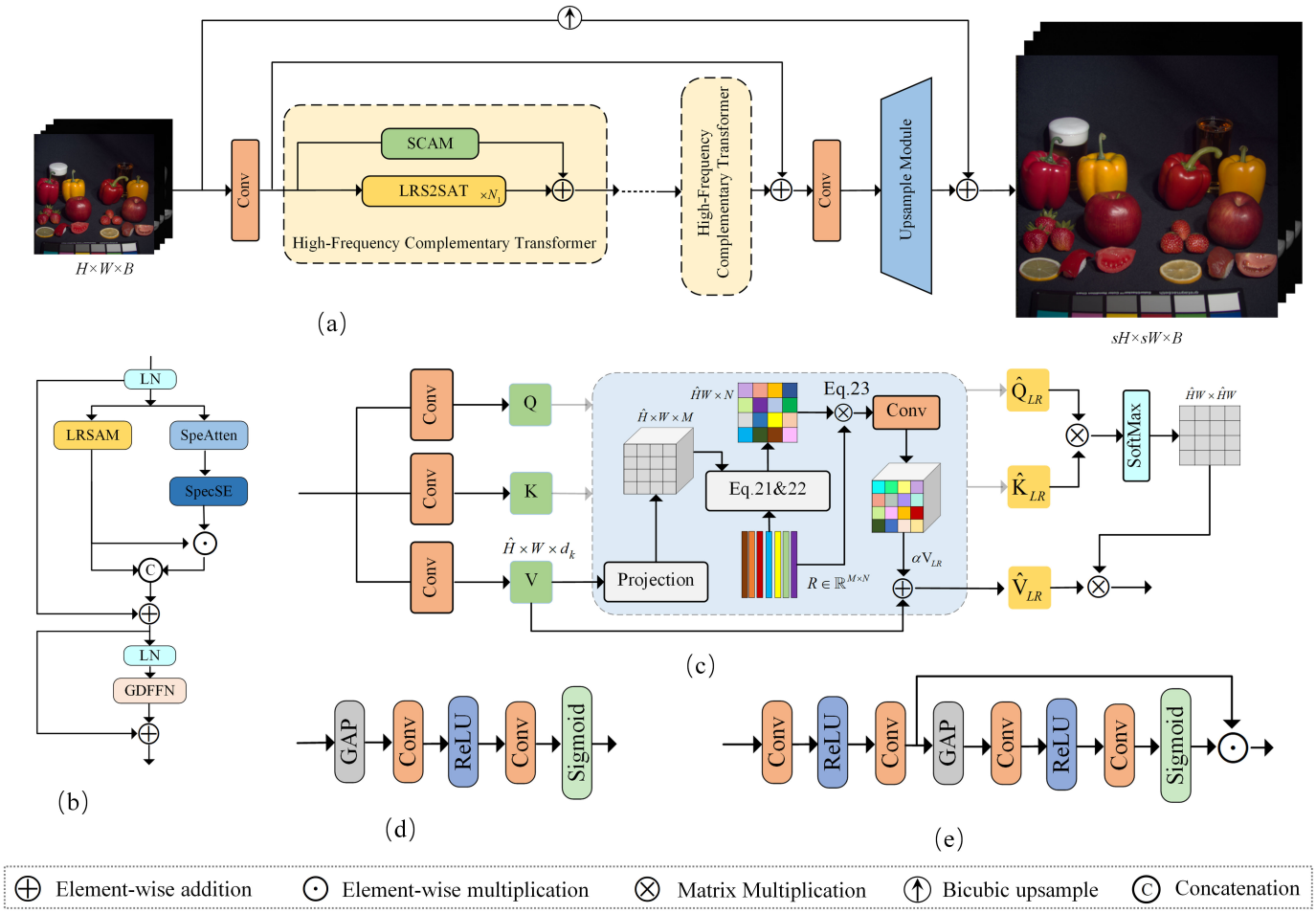


Fig. 2. Overview of our approach. (a) Main architecture of the proposed method. (b) Proposed LRS2SAT. (c) LRSAM. (d) Spectral squeeze-and-excitation module. (e) SCAM.

to hard one-hot basis selection, rather than merely a generic nonlinear activation. This allows the model to preserve the preference for dominant bases while avoiding the discontinuity and instability of hard assignments, thereby yielding a more robust and trainable representation.

IV. Proposed Method

In this section, we describe the proposed method in detail, including the overall architecture, the LRS2SAT block, the LRSAM, the spectral squeeze-and-excitation module, the SCAM, and the loss functions.

A. Overall Architecture

Our method exploits the inherent spectral low-rank structure of HSIs while enabling effective spectral-to-spatial information integration. As shown in Fig. 2, the overall architecture consists of three components: shallow feature extraction, deep feature extraction, and image reconstruction.

Given a LR HSI $I_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B denote the spatial height, width, and the number of spectral bands, respectively, the proposed model produces a super-resolved HSI $I_{\mathrm{SR}} \in \mathbb{R}^{sH\times sW\times B}$, where s is the scale factor. Specifically, we learn an end-to-end mapping $\mathcal{F}_{\mathrm{LRS2ST}}(\cdot)$ from I_{LR} to I_{SR}

$$I_{\mathrm{SR}} = \mathcal{F}_{\mathrm{LRS2ST}}\left(I_{\mathrm{LR}}\right) \tag{8}$$

where $\mathcal{F}_{\mathrm{LRS2ST}}(\cdot)$ denotes the proposed SR network.

Specifically, the proposed LRS2ST first extracts shallow features from the input I_{LR}) using a 3×3 convolution

$$F_0 = \mathcal{F}_{\mathrm{s}}\left(I_{\mathrm{LR}}\right) \tag{9}$$

where $\mathcal{F}_s(\cdot)$ denotes the shallow feature extraction layer and F_0 is the resulting feature map. The shallow features are then fed into the deep feature extraction module, which consists of N HFCT blocks. The output of the nth block is given by

$$F_n = \mathcal{B}_n\left(F_{n-1}\right), \quad n = 1, 2, \ldots, N \tag{10}$$

where $\mathcal{B}_n(\cdot)$ denotes the mapping of the nth HFCT block and F_n is the corresponding output feature. To aggregate shallow and deep representations, we combine F_{N} with F_0 and apply a convolution to obtain the enhanced feature F_{d}

$$F_{\mathrm{d}} = \mathcal{F}_{\mathrm{conv}}\left(F_{\mathrm{N}} + F_0\right) \tag{11}$$

where $\mathcal{F}_{\mathrm{conv}}(\cdot)$ denotes a convolutional layer. In the reconstruction stage, the super-resolved HSI is obtained via an upsampling module with a global skip connection

$$I_{\mathrm{SR}} = \mathcal{F}_{\mathrm{upsample}}\left(F_{\mathrm{d}}\right) + I_{\mathrm{LR}}^{\mathrm{bic}} \tag{12}$$

where $I_{\mathrm{LR}}^{\mathrm{bic}}$ is the bicubically upsampled version of LR, and $\mathcal{F}_{\mathrm{upsample}}(\cdot)$ denotes the upsampling module.

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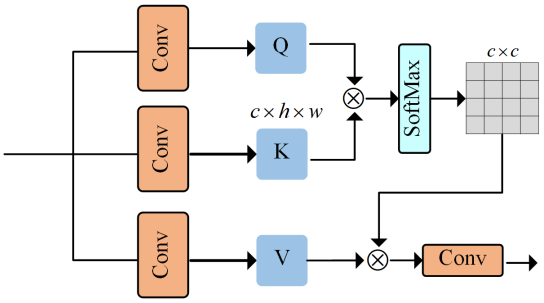


Fig. 3. Spectral attention module.

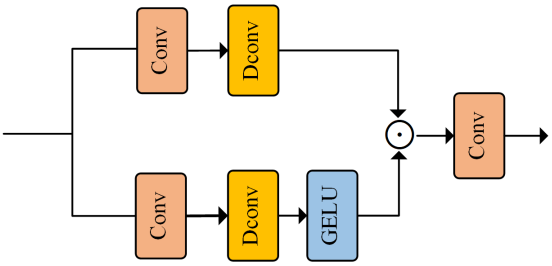


Fig. 4. Structure of proposed GDFNN.

B. HFCT

As reported in [35], Transformer architectures are effective at modeling low-frequency information but are less capable of capturing high-frequency details than CNNs. To alleviate this limitation, we propose the HFCT, which combines an LRS2SAT branch with an SCAM to enhance high-frequency feature representation, as shown in Fig. 2(a).

The LRS2SAT is designed to effectively exploit the spectral low-rank characteristics of HSIs and to guide spatial feature modulation using spectral attention weights, which are extracted by the SpecSE module to reflect the band-wise importance and thereby enhance the spatial feature representations. As shown in Fig. 2(b), given an input feature $F \in \mathbb{R}^{H\times W\times C}$, the feature is first processed by a LayerNorm (LN) layer, followed by two parallel branches: a spatial attention module and a spectral attention module. After these modules, the spectral attention features are passed to the SpecSE module to produce an attention map, which is subsequently used to modulate the spatial attention features via element-wise multiplication. The modulated features are subsequently concatenated with the original spatial attention features and passed through an LN layer, followed by a gated depthwise convolutional feed-forward network (GDFFN). The overall process can be formulated as follows:

$$\begin{aligned} F_{\text{spa}} &= \text{LRSAM}\left(\text{LN}\left(F\right)\right) \\ F_{\text{spe}} &= \text{SpeAtten}\left(\text{LN}\left(F\right)\right) \\ \tilde{F}_{\text{spa}} &= \text{SpecSE}\left(F_{\text{spe}}\right) \odot F_{\text{spa}} \\ F_{\text{cat}} &= \text{Concat}\left(F_{\text{spa}}, \tilde{F}_{\text{spa}}\right) \\ F_{\text{fusion}} &= \text{Conv}_{1\times 1}\left(F_{\text{cat}}\right) + F \\ F_{\text{out}} &= \text{GDFFN}\left(\text{LN}\left(F_{\text{fusion}}\right)\right) + F_{\text{fusion}} \end{aligned} \tag{13}$$

where $\odot$ denotes the element-wise multiplication and Concat$(\cdot,\cdot)$ denotes the channel-wise concatenation. Here, F_{spa}, $\tilde{F}_{\text{spa}}$, F_{spe}, F_{fusion}, and F_{out} are all of size $H \times W \times C$, while F_{cat} is of size $H \times W \times 2C$. The operator $\text{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolution used to project the concatenated features back to C channels, and LRSAM$(\cdot)$, SpeAtten$(\cdot)$, LN$(\cdot)$, SpecSE$(\cdot)$, and GDFFN$(\cdot)$ denote the LRSAM, spectral attention module, layer normalization, SpecSE module, and GDFFN, respectively.

The LRSAM is illustrated in Fig. 2(c), and a detailed explanation will be provided in Section IV-C. The spectral attention module is depicted in Fig. 3. Specifically, a 1×1 convolution is first applied to extract the query, key, and value features, denoted as Q, K, and $V \in \mathbb{R}^{c\times h\times w}$, respectively. We then employ the multihead attention mechanism; for clarity, we describe the single-head case as an example, which can be formulated as follows:

$$F_{\text{spe}} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{14}$$

where d is the feature dimension, and Softmax$(\cdot)$ represents the softmax function.

As illustrated in Fig. 2(d), the SpecSE is formulated as

$$\text{SpecSE}\left(F_{\text{spe}}\right) = f\left(W^2_{\text{spe}}\text{ReLU}\left(W^1_{\text{spe}}\text{GAP}\left(F_{\text{spe}}\right)\right)\right)$$

where $f(\cdot)$ denotes the sigmoid function, ReLU$(\cdot)$ represents the ReLU activation, GAP$(\cdot)$ is the global average pooling operation, and $W^{(\cdot)}_{\text{spe}}$ are the 1×1 convolutional layers.

The GDFFN, illustrated in Fig. 4, follows the Gated-Dconv Feed-Forward Network (GDFN) in Restormer [52]. The process can be formulated as

$$\begin{aligned} F_1 &= W^1_d\left(W^1_p F\right) \\ F_2 &= \text{GELU}\left(W^2_d\left(W^2_p F\right)\right) \\ \text{GDFNN}\left(F\right) &= F_1 \odot F_2 \end{aligned} \tag{15}$$

where W^i_d and W^i_p denote the point-wise convolution layers and depth-wise convolution layers, respectively, and GELU$(\cdot)$ represents the GELU activation function.

To enhance the high-frequency feature representation, we incorporate the SCAM into the Transformer block. As shown in Fig. 2(d), SCAM captures both spatial and channel dependencies by applying convolutions and GAP. This process can be denoted as

$$\begin{aligned} F_{\text{middle}} &= W^1\text{ReLU}\left(W^2 F\right) \\ \text{Atten} &= f\left(\text{GAP}\left(W^3\text{ReLU}\left(W^4 F_{\text{middle}}\right)\right)\right) \\ \text{SCAM}\left(F\right) &= \text{Atten} \odot F_{\text{middle}} \end{aligned} \tag{16}$$

where F_{middle} and Atten represent the middle feature and attention map, respectively, $\odot$ represents the element-wise multiplication, $W^i (i = 1, 2, 3, 4)$ is the convolution layer, GAP$(\cdot)$ is the global average pooling, and $f(\cdot)$ denotes the sigmoid function. The attention map generated by SCAM enables the model to emphasize critical high-frequency components, thereby improving its ability to represent fine-grained details in HSIs.

The proposed HFCT module facilitates efficient interaction between the spatial and spectral domains while effectively exploiting the low-rank characteristics of HSIs. In addition, it

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strengthens the extraction of high-frequency details, to which Transformers are typically less sensitive.

C. LRSAM

The details of LRSAM are illustrated in Fig. 2(c), where the proposed LRR is integrated. Specifically, we adopt the cross-shaped window (CSWin) Transformer [53], which performs self-attention in horizontal and vertical stripes in parallel. Let $F \in \mathbb{R}^{H\times W\times C}$ denote the input feature of LRSAM. We first split F along the channel dimension into two parts

$$F = [X_1, X_2] \tag{17}$$

$$X_1, X_2 \in \mathbb{R}^{H\times W\times C/2}. \tag{18}$$

Different from [19] and [54], we perform both horizontal- and vertical-stripe self-attention on X_1 and X_2.

For X_1, which is processed by horizontal-stripe self-attention, we evenly partition X_1 into N nonoverlapping horizontal stripes $[X_1^1, \ldots, X_1^N]$, each of size $\hat{H} \times W \times (C/2)$, where $\hat{H} = H/N$. For the kth attention head, we define

$$Q_k^i = X_1^i W_k^q, K_k^i = X_1^i W_k^k, V_k^i = X_1^i W_k^v \tag{19}$$

$$Y_k^i = \text{LRSAM}\left(Q_k^i, K_k^i, V_k^i\right) \tag{20}$$

$$\text{H-Attention}_k\left(X_1\right) = \left[Y_k^1, Y_k^2, \ldots, Y_k^N\right] \tag{21}$$

where $W_k^q, W_k^k, W_k^v \in \mathbb{R}^{(C/2)\times d_k}$ are the projection matrices for queries, keys, and values of the kth head, respectively, and d_k is set to $(C/2)/K$. The vertical-stripe self-attention V-Attention$_k(\cdot)$ for the kth head is obtained in an analogous manner. Following [53], the K attention heads are divided into two parallel groups, each containing $K/2$ heads (we assume K is even). The first group performs self-attention on horizontal stripes, while the second group operates on vertical stripes. The final output of the multihead attention is obtained by concatenating the results of the two groups along the channel dimension. In contrast to X_1, for X_2, the first group performs self-attention on vertical stripes, whereas the second group performs self-attention on horizontal stripes.

In the proposed LRSAM framework, to effectively capture both spectral low-rankness and spatial self-similarity, we incorporate LRR into the attention mechanism. As illustrated in Fig. 2(c), we take the value tensor $V \in \mathbb{R}^{\hat{H}\times W\times d_k}$ as an example. First, V is passed through a 1×1 convolutional layer, yielding a transformed feature map $\tilde{V} \in \mathbb{R}^{\hat{H}\times W\times M}$, where M denotes the output channel dimension. For simplicity, we still denote this tensor by V. Here, M also represents the dimensionality of each basis vector in the subsequent LRR process. We then reshape V into a 2-D matrix $V \in \mathbb{R}^{\hat{H}W\times M}$, where each row corresponds to a spatial location and each column corresponds to a feature dimension. The LRR basis matrix is defined as $R \in \mathbb{R}^{M\times N}$, where N is the number of basis vectors used in the representation.

For the feature vector at spatial location (h, w), denoted by $V_{hw} \in \mathbb{R}^{1\times M}$, its response with the nth low-rank basis vector $R_n \in \mathbb{R}^{M\times 1}$ is quantified using the dot product

$$c_{hw,n} = \sum_{m=1}^{M} V_{hw,m} R_{m,n} \tag{22}$$

where $R_{m,n}$ denotes the mth element of the nth basis vector. The resulting scalar $c_{hw,n}$ is a response score that reflects how well the input feature V_{hw} matches the nth basis vector R_n.

The resulting scalar $c_{hw,n}$ is referred to as the representation coefficient, which indicates the contribution of the nth basis vector R_n to the reconstruction of the feature vector V_{hw}.

To introduce nonlinearity into the LRR and enhance its compactness and stability, we apply a softmax operation to $c_{hw,n}$ along the basis dimension

$$S_{hw,n} = \frac{\exp\left(c_{hw,n}\right)}{\sum_{j=1}^{N} \exp\left(c_{hw,j}\right)} \tag{23}$$

where $S_{hw,n}$ denotes the normalized response score of the nth basis vector with respect to the input feature at position (h, w). Since the softmax is applied independently at each spatial position (h, w), the importance scores vary across different locations. As a result, spatially similar regions are more likely to activate similar dominant bases and thus exhibit similar basis-assignment patterns, which increases the row-wise correlation of S and facilitates a more compact LRR. Since texture patterns often exhibit strong spatial redundancy and self-similarity, such structured low-rank modeling is also beneficial for preserving textural consistency and fine spatial details.

Using these normalized weights, the reconstructed low-rank feature $V_{\text{LR}} \in \mathbb{R}^{\hat{H}W\times M}$ is computed as

$$\left(V_{\text{LR}}\right)_{hw,m} = \sum_{n=1}^{N} S_{hw,n} \left(R^{\text{T}}\right)_{n,m} \tag{24}$$

where $\left(R^{\text{T}}\right)_{n,m} = R_{m,n}$ denotes the mth element of the transposed nth basis vector. The entire LRR forward process can be compactly expressed as

$$V_{\text{LR}} = \text{Softmax}\left(VR\right) R^{\text{T}} \tag{25}$$

which yields a nonlinear and input-adaptive low-rank reconstruction while relaxing the conventional fixed linear projection used in low-rank modeling.

After obtaining V_{LR}, we reshape it into a tensor of size $\hat{H} \times W \times M$. Subsequently, a 1×1 convolutional layer is applied to project it to $\mathbb{R}^{\hat{H}\times W\times d_k}$. For notational simplicity, we still denote the resulting tensor by V_{LR}. Finally, we obtain the enhanced representation $\hat{V}_{\text{LR}}$ via a residual, element-wise fusion of the low-rank feature V_{LR} with the original feature V

$$\hat{V}_{\text{LR}} = \alpha V_{\text{LR}} + V \tag{26}$$

where α is a scaling coefficient. This residual formulation allows the model to introduce a low-rank prior through V_{LR} while preserving fine-grained details from the original features via the residual term V. The coefficient α balances these two contributions, thereby achieving a better trade-off between robustness and representational capacity. Similarly, the low-rank query and key features $\hat{Q}_{\text{LR}}$ and $\hat{K}_{\text{LR}}$ are obtained and used in LRSAM as

$$Y = \text{Softmax}\left(\frac{\hat{Q}_{\text{LR}} \hat{K}_{\text{LR}}^{\top}}{\sqrt{d}}\right) \hat{V}_{\text{LR}} \tag{27}$$

where d is the feature dimension used for scaling.

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By leveraging the inherent spectral dependencies of HSIs, the proposed LRR strategy produces compact low-dimensional representations that preserve spectral integrity and effectively capture the underlying data structure.

D. Loss Function

In HSI SR tasks that leverage fine-grained 3-D information, ensuring both spatial fidelity and spectral consistency remains a significant challenge. Among various optimization strategies, the $\mathcal{L}_1$ loss is widely adopted, as it effectively preserves fine structural details and sharp boundaries. The loss function is formulated as

$$\mathcal{L}_1(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\| I_{\mathrm{SR}}^n - I_{\mathrm{HR}}^n \right\|_1 \tag{28}$$

where N is the number of images in each training batch, I_{SR}^n and I_{HR}^n represent the nth reconstructed HSI and the corresponding HR reference, respectively.

To quantitatively evaluate the spectral similarity between the reconstructed HSI and its corresponding HR counterpart, the spectral angle mapper (SAM) loss [18] is employed, which is defined as

$$\mathcal{L}_{sam}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\arccos\left(\frac{I_{\mathrm{SR}}^n \cdot I_{\mathrm{HR}}^n}{\left\| I_{\mathrm{SR}}^n \right\|_2 \cdot \left\| I_{\mathrm{HR}}^n \right\|_2}\right). \tag{29}$$

The SAM loss explicitly accounts for the spectral characteristics of HSIs, thereby compensating for the limitations of traditional pixel-wise loss functions.

We also introduce a gradient loss to enhance the sharpness of the reconstructed images. By computing the difference between neighboring pixels, we can obtain the gradient maps of HSI I_{SR}, which can be denoted as

$$\nabla I_{\mathrm{SR}} = (\nabla_h I_{\mathrm{SR}}, \nabla_w I_{\mathrm{SR}}, \nabla_s I_{\mathrm{SR}}) \tag{30}$$

$$M(I_{\mathrm{SR}}) = \|\nabla I_{\mathrm{SR}}\|_2 \tag{31}$$

where the $\nabla_h I_{\mathrm{SR}}$, $\nabla_w I_{\mathrm{SR}}$, and $\nabla_s I_{\mathrm{SR}}$ refer to the gradient value along horizontal, vertical, and spectral dimensions of I_{SR}, respectively. Then the gradient loss can be formulated as follows:

$$\mathcal{L}_{gra}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\| M\left(I_{\mathrm{SR}}^n\right) - M\left(I_{\mathrm{HR}}^n\right) \right\|_1. \tag{32}$$

Then the total loss for training can be denoted as follows:

$$\mathcal{L}_{total}(\Theta) = \mathcal{L}_1 + \lambda_1 \mathcal{L}_{sam} + \lambda_2 \mathcal{L}_{gra} \tag{33}$$

where λ_1 and λ_2 denote the hyperparameters that control the weight of SAM loss and gradient loss, respectively.

V. Experiments

A. Datasets

We firstly introduce three widely used benchmark HSI datasets, i.e., CAVE, Pavia Center, and Chikusei.

1) CAVE Dataset: The CAVE dataset, which was captured using a cooled CCD camera with wavelengths ranging from 400 to 700 nm. The dataset consists of 32 HSIs, each with a spatial size of 512 $\times$ 512 and 31 spectral bands.

2) Pavia Center Dataset: The Pavia Center dataset was acquired by the ROSIS sensor during a flight campaign over the center area of Pavia, northern Italy. It has a spatial size of 1096 $\times$ 1096 and contains 102 spectral bands covering the wavelength range from 430 to 860 nm, and the ground sampling distance is 1.3 m.

3) Chikusei Dataset: The Chikusei dataset was acquired using the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban regions in Chikusei, Japan. It covers a spectral range of 363–1018 nm and consists of 128 bands with a spatial resolution of 2517 $\times$ 2335 pixels.

B. Implementation Details

For the CAVE dataset, 80% of the data was randomly allocated for training, 10% for validation, and the remaining 10% for testing. For the Pavia Center dataset, due to the invalid region, following [42], we crop the informative region with 1096 $\times$ 715 $\times$ 102 of the original scene. We crop the top regions of the image into four nonoverlapping images with the size of 128 $\times$ 128 $\times$ 102 for testing. The rest of the region of the image is cropped to form the overlap patches for training (10% of the training data is randomly selected as a validation set). For the Chikusei dataset, similar to the setting of the Pavia Center dataset, the images were partitioned into four nonoverlapping top regions, each with a size of 512 $\times$ 512 $\times$ 128, to form the test set. Additionally, overlapping patches were extracted from the remaining regions to construct the training set, with 10% of the training data held out for validation. To ensure experimental consistency and enable fair comparison with existing methods, these images are further bicubically downsampled by factors of $\times 2$, $\times 4$, or $\times 8$ to generate LR images for constructing LR–HR image pairs.

In the proposed method, both the standard convolution and depth-wise convolution employ a kernel size of 3 $\times$ 3. The number of feature channels is set to 180. Additionally, the number of HFCT and LRSAM modules is configured to 4 each. It is noteworthy that our method performs SR progressively by PixelShuffle [55] to reduce reconstruction difficulty. Specifically, for a scale factor of 4, two consecutive $\times 2$ SR operations are applied; for a scale factor of 8, three consecutive $\times 2$ SR operations are conducted. The hyperparameters λ_1 and λ_2 in the loss function are set to 0.3 and 0.1, respectively. The Adam optimizer was used to train the proposed network for 200 epochs. The initial learning rate was set to 1×10^{-4} and was halved every 100 epochs, up to a total of 200 epochs. The batch size is set as 32. All experiments are implemented by the PyTorch framework on NVIDIA RTX 3090 GPUs.

To evaluate the effectiveness of the proposed method, we conducted comparative experiments against several state-of-the-art approaches, including GDRRN [18], SSPSR [41], EUNet [17], AS3UNet [20], MSDformer [42], ESSAformer [23], SRDNet [22], CST [19], and HiT-SRF [7]. Among these, GDRRN, SSPSR, EUNet, and SRDNet are CNN-based methods, while AS3UNet, MSDformer, ESSAformer, CST, and HiT-SRF are Transformer-based methods. For all comparative models, we made efforts to achieve their optimal performance for a fair evaluation. To demonstrate the performance

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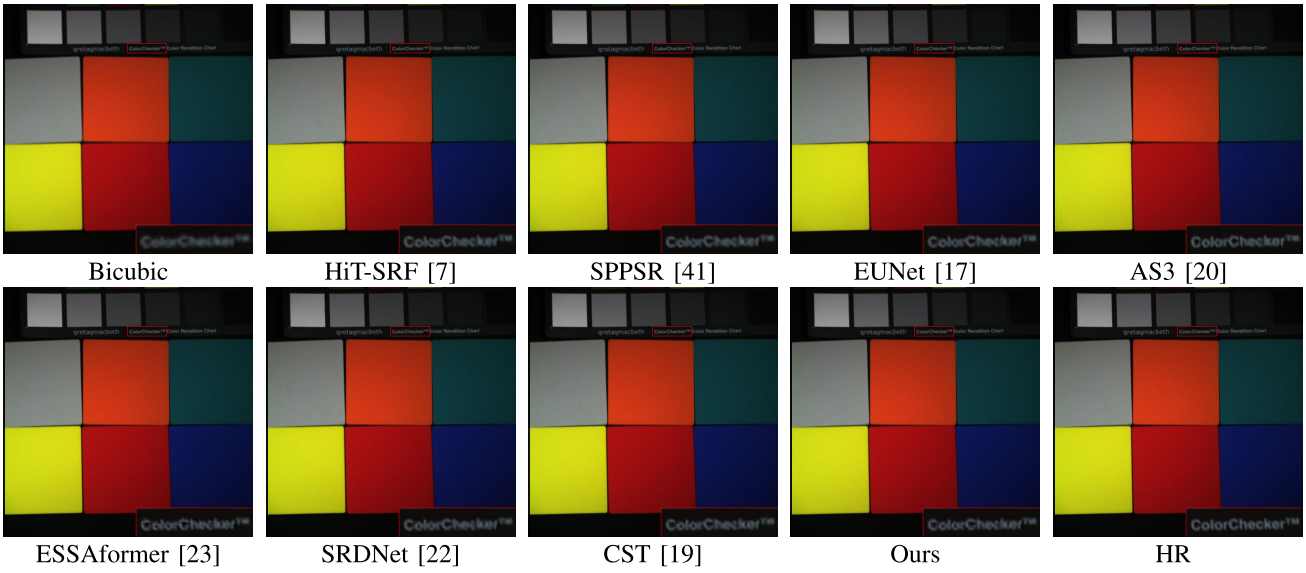
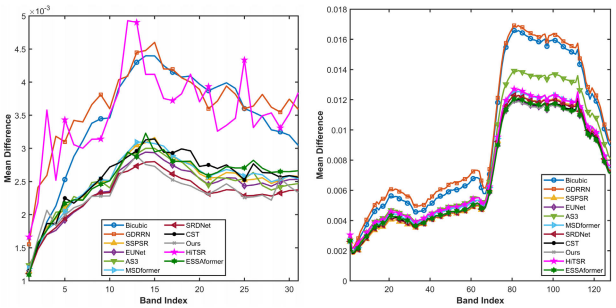


Fig. 5. Reconstructed sponges_ms image of different methods from the CAVE dataset, using spectral bands 23rd, 15th, and 8th as the R, G, and B channels, with a $\times 2$ upsampling factor.



(a) (b)

Fig. 6. Mean spectral difference curves of different methods. (a) sponges_ms image from CAVE dataset. (b) Test image from the Chikusei dataset.

of the models in both spatial and spectral dimensions, six widely used quantitative image quality metrics were employed: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), SAM, cross correlation (CC), root-mean-squared error (RMSE), and erreur relative globale adimensionnelle de synthese (ERGAS).

C. Experiments on the CAVE Dataset

Table I presents the quantitative performance of the proposed method compared with existing approaches on the CAVE dataset at scale factors of $\times 2$, $\times 4$, and $\times 8$. The best and second-best results are highlighted in red and blue, respectively. As shown in the results, the proposed method consistently achieves the highest PSNR, SSIM, and CC values, along with the lowest SAM, RMSE, and ERGAS values across almost all scale factors. These results clearly demonstrate the superior ability of the proposed method to preserve both spectral and spatial information. To further validate the effectiveness of the proposed method, Fig. 5 provides a visual comparison of reconstruction results. For improved visualization, the 23rd, 15th, and 8th spectral bands are combined to form pseudo-RGB composite images. The proposed method recovers fine image details, such as textual structures, more effectively than the compared methods, which often yield

TABLE I

QUANTITATIVE COMPARISON FOR DIFFERENT METHODS ON THE CAVE DATASET. THE BEST AND SECOND-BEST PERFORMING RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
Bicubic		40.148	0.9786	3.059	0.9904	0.0102	15.622
GDRRN [18]		40.733	0.9803	3.299	0.9902	0.0102	16.330
SSPSR [41]		43.477	0.9861	2.747	0.9921	0.0079	15.634
EUNet [17]		43.849	0.9871	2.536	0.9931	0.0078	16.266
AS3UNet [20]	$\times 2$	44.086	0.9873	2.491	0.9932	0.0076	14.010
MSDformer [42]		43.427	0.9863	2.634	0.9927	0.0073	15.774
ESSAformer [23]		44.157	0.9877	2.555	0.9932	0.0067	11.106
SRDNet [22]		44.058	0.9868	2.640	0.9934	0.0075	10.894
CST [19]		44.085	0.9865	2.451	0.9985	0.0072	14.208
HiT-SRF [7]		42.336	0.9849	2.954	0.9920	0.0081	15.368
Ours		44.371	0.9887	2.412	0.9988	0.0065	10.982
Bicubic		33.956	0.9315	4.194	0.9781	0.0216	14.201
GDRRN [18]		34.247	0.9342	4.744	0.9797	0.0200	9.087
SSPSR [41]		37.932	0.9586	3.432	0.9862	0.0141	8.475
EUNet [17]		37.235	0.9563	3.433	0.9852	0.0157	8.750
AS3UNet [20]	$\times 4$	37.948	0.9588	3.346	0.9866	0.0148	7.552
MSDformer [42]		37.194	0.9549	3.424	0.9853	0.0151	8.623
ESSAformer [23]		37.990	0.9604	3.579	0.9858	0.0145	7.657
SRDNet [22]		37.622	0.9574	3.507	0.9861	0.0154	7.515
CST [19]		37.980	0.9611	3.362	0.9943	0.0145	7.394
HiT-SRF [7]		36.632	0.9523	3.950	0.9846	0.0159	8.343
Ours		38.234	0.9614	3.344	0.9947	0.0141	7.057
Bicubic		29.523	0.8603	5.796	0.9587	0.0349	21.818
GDRRN [18]		29.235	0.8626	6.500	0.9569	0.0356	18.190
SSPSR [41]		32.453	0.9057	4.542	0.9750	0.0256	9.030
EUNet [17]		32.530	0.9049	4.407	0.9740	0.0254	8.789
AS3UNet [20]	$\times 8$	32.967	0.9102	4.311	0.9756	0.0247	7.480
MSDformer [42]		31.899	0.8984	4.368	0.9740	0.0272	13.202
ESSAformer [23]		32.960	0.9100	4.830	0.9760	0.0245	7.898
SRDNet [22]		32.715	0.9057	4.694	0.9747	0.0251	10.800
CST [19]		32.680	0.9068	4.233	0.9760	0.0252	8.975
HiT-SRF [7]		31.819	0.8966	5.429	0.9726	0.0273	13.309
Ours		33.076	0.9120	4.215	0.9763	0.0243	7.211

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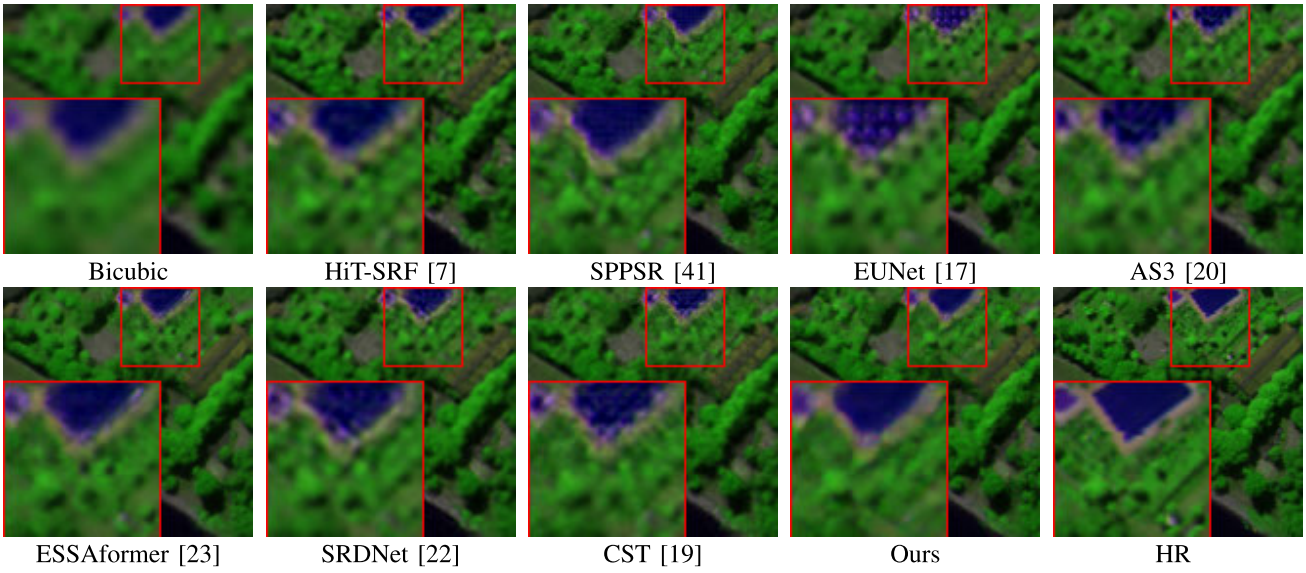


Fig. 7. Reconstructed image of different methods from the Pavia Center dataset, using spectral bands 70th, 100th, and 36th as the R, G, and B channels, with a $\times 4$ upsampling factor.

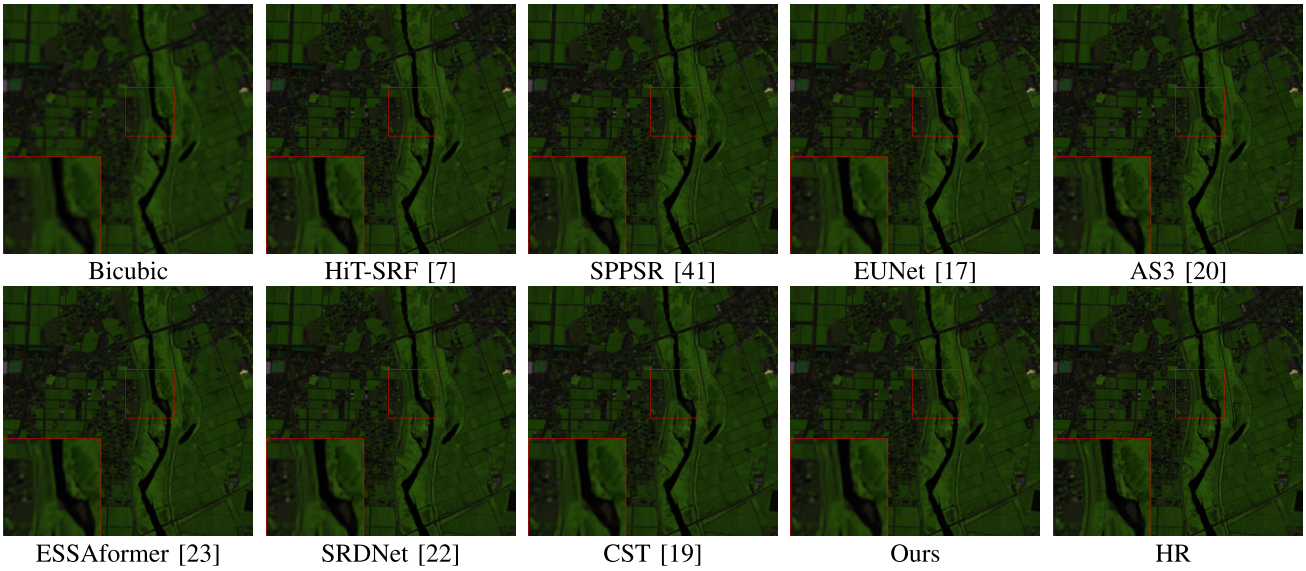


Fig. 8. Reconstructed image of different methods from the Chikusei dataset, using spectral bands 70th, 100th, and 36th as the R, G, and B channels, with a $\times 4$ upsampling factor.

blurred and indistinct results. In terms of spectral fidelity, the mean spectral difference curves of various methods are illustrated in Fig. 6(a). Among all methods, the proposed method exhibits the lowest curve, indicating the highest spectral accuracy. Overall, the experimental results demonstrate that the proposed method effectively preserves both spatial and spectral information, achieving superior performance in HSI SR.

D. Experiments on the Pavia Center Dataset

Table II presents the quantitative comparison of seven HSI SR methods on the Pavia Center dataset at scale factors of $\times 2$, $\times 4$, and $\times 8$. As shown in the results, the proposed method delivers the most competitive overall performance. By jointly exploiting spectral and spatial information, it demonstrates superior effectiveness in preserving spatial details and maintaining spectral consistency. Fig. 7 illustrates the qualitative results of all compared methods at a scale factor of $\times 4$ on the Pavia Center dataset. A pseudo-color composite image is produced by assigning the 70th, 100th, and 36th spectral bands to the R, G, and B channels, respectively. Early approaches, such as GDRRN [18], SSPSR [41], and EUNet [17], often produce artifacts and exhibit noticeable blurriness in the reconstructed images. Although more recent methods improve visual quality, they still struggle to recover clear and sharp edges. Moreover, while HiT-SRF [7] has shown strong performance on natural RGB images, its direct application to HSI SR delivers only limited gains. In contrast, the proposed method better preserves fine textures and edge details, resulting in reconstructions with higher visual fidelity and structural accuracy. This superior performance can be largely attributed

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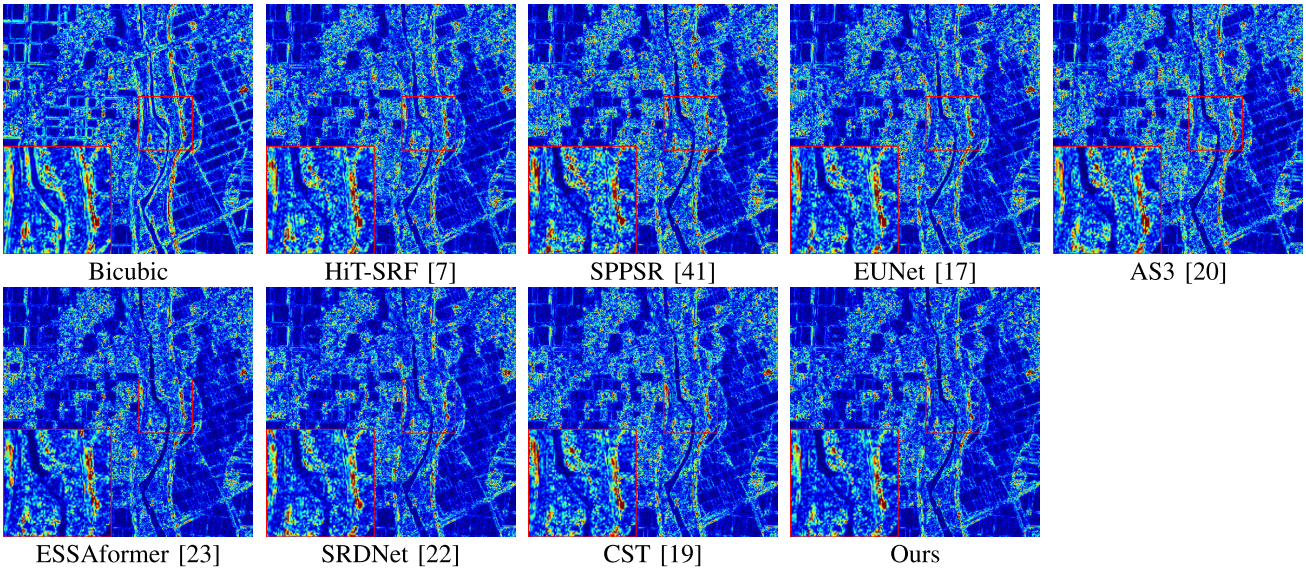


Fig. 9. Error maps of the test HSI on the Chikusei dataset at $\times 4$ upsampling factor.

TABLE II

QUANTITATIVE COMPARISON FOR DIFFERENT METHODS ON THE PAVIA CENTER DATASET. THE BEST AND SECOND-BEST PERFORMING RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
Bicubic		36.386	0.9378	5.677	0.9791	0.0179	8.344
GDRRN [18]		36.457	0.9453	5.380	0.9801	0.0171	8.080
SSPSR [41]		39.067	0.9645	5.187	0.9893	0.0127	6.230
EUNet [17]		39.251	0.9656	5.172	0.9898	0.0121	6.162
AS3UNet [20]	$\times 2$	38.558	0.9601	5.219	0.9878	0.0134	6.490
MSDformer [42]		38.828	0.9622	5.342	0.9884	0.0127	6.432
ESSAformer [23]		39.480	0.9655	5.152	0.9903	0.0121	6.044
SRDNet [22]		39.005	0.9646	5.197	0.9884	0.0123	6.294
CST [19]		39.253	0.9654	5.233	0.9890	0.0120	6.170
HiT-SRF [7]		38.769	0.9606	5.774	0.9873	0.0127	6.572
Ours		39.544	0.9670	5.132	0.9945	0.0116	5.972
Bicubic		31.524	0.8223	7.272	0.9401	0.0306	7.133
GDRRN [18]		31.796	0.8394	7.174	0.9434	0.0302	7.058
SSPSR [41]		32.972	0.8683	6.917	0.9560	0.0260	6.028
EUNet [17]		32.641	0.8643	7.062	0.9524	0.0274	6.305
AS3UNet [20]	$\times 4$	32.232	0.8526	7.099	0.9485	0.0285	6.491
MSDformer [42]		32.648	0.8547	7.402	0.9529	0.0269	6.245
ESSAformer [23]		33.143	0.8731	6.703	0.9575	0.0253	5.922
SRDNet [22]		32.945	0.8672	6.884	0.9554	0.0262	6.088
CST [19]		33.044	0.8729	6.681	0.9592	0.0259	5.998
HiT-SRF [7]		32.682	0.8622	7.452	0.9531	0.0265	6.253
Ours		33.355	0.8799	6.491	0.9599	0.0252	5.837
Bicubic		27.889	0.6672	8.842	0.8634	0.0438	5.192
GDRRN [18]		28.001	0.6701	8.894	0.8659	0.0445	5.201
SSPSR [41]		28.291	0.7116	8.599	0.8767	0.0434	5.100
EUNet [17]		28.362	0.6894	9.707	0.8773	0.0428	5.037
AS3UNet [20]	$\times 8$	28.234	0.6963	9.369	0.8776	0.0440	5.070
MSDformer [42]		28.267	0.6891	9.407	0.8756	0.0445	5.103
ESSAformer [23]		28.483	0.6968	9.282	0.8801	0.0416	5.059
SRDNet [22]		28.358	0.6962	9.357	0.8777	0.0430	5.102
CST [19]		28.314	0.6770	9.365	0.8755	0.0427	5.125
HiT-SRF [7]		28.409	0.6964	10.518	0.8779	0.0420	5.053
Ours		28.613	0.6970	9.141	0.8818	0.0416	4.973

to the proposed nonlinear LRR module, which enables more compact and adaptive modeling of spatially varying features.

E. Experiments on the Chikusei Dataset

The quantitative and qualitative results on the chikusei dataset are presented in Table III and Fig. 8, respectively. As shown in Table III, we evaluate the compared methods at scale factors of $\times 2$, $\times 4$, and $\times 8$. The proposed method consistently delivers strong performance across different scaling factors, achieving the best results across all six evaluation metrics at $\times 2$ and maintaining competitive performance at $\times 4$ and $\times 8$. In particular, the proposed method consistently achieves the lowest SAM values across all scale levels, demonstrating its strong ability to preserve spectral fidelity. Furthermore, Fig. 8 shows the visual reconstruction of a representative HSI from the Chikusei test set at a scale factor of $\times 4$. For visualization, the 70th, 100th, and 36th spectral bands are mapped to the R, G, and B channels, respectively, to generate a pseudo-color composite image. Compared with other methods, the proposed approach produces clearer edge structures, especially within the red-marked region. To further evaluate pixel-wise reconstruction accuracy, the mean spectral error maps are visualized in Fig. 9. These maps provide an intuitive overview of reconstruction performance across the entire image, where regions shaded in blue indicate higher fidelity. As highlighted in the red box, the proposed method yields noticeably lower error levels than the compared approaches, underscoring its effectiveness in recovering fine details accurately. Moreover, Fig. 6(b). shows the mean spectral difference curves of the reconstructed results at a scale factor of $\times 4$ on the Chikusei test dataset. These curves provide a spectral-domain evaluation of SR performance, where a lower curve indicates smaller discrepancies from the ground-truth spectra. As shown in the results, the proposed method consistently attains the lowest spectral deviations, demonstrating its effectiveness in preserving spectral integrity.

F. Ablation Study

In this section, we conduct the ablation experiments on the CAVE dataset for $\times 2$ scale.

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TABLE III

QUANTITATIVE COMPARISON FOR DIFFERENT METHODS ON THE CHIKUSEI DATASET. THE BEST AND SECOND-BEST PERFORMING RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
Bicubic		41.832	0.9701	1.788	0.9923	0.0082	4.990
GDRRN [18]		41.955	0.9739	1.876	0.9926	0.0081	4.903
SSPSR [41]		41.162	0.9644	3.046	0.9914	0.0088	5.340
EUNet [17]		45.459	0.9869	1.245	0.9967	0.0054	3.284
AS3UNet [20]	×2	44.411	0.9837	1.434	0.9958	0.0061	3.700
MSDformer [42]		45.227	0.9861	1.284	0.9965	0.0055	3.368
ESSAformer [23]		45.757	0.9874	1.344	0.9967	0.0052	3.133
SRDNet [22]		45.521	0.9869	1.288	0.9967	0.0054	3.245
CST [19]		45.578	0.9872	1.274	0.9968	0.0053	3.239
HiT-SRF [7]		45.152	0.9859	1.544	0.9964	0.0056	3.402
Ours		45.907	0.9882	1.204	0.9970	0.0051	3.116
Bicubic		36.246	0.8951	3.404	0.9720	0.0169	4.737
GDRRN [18]		36.294	0.8948	3.332	0.9683	0.0166	4.535
SSPSR [41]		37.993	0.9297	2.541	0.9806	0.0128	3.901
EUNet [17]		38.162	0.9310	2.594	0.9813	0.0126	3.825
AS3UNet [20]	×4	37.852	0.9229	2.719	0.9805	0.0139	3.940
MSDformer [42]		37.831	0.9256	2.759	0.9798	0.0130	3.975
ESSAformer [23]		38.205	0.9307	2.666	0.9819	0.0130	3.858
SRDNet [22]		37.986	0.9286	2.658	0.9805	0.0128	3.906
CST [19]		38.200	0.9308	2.637	0.9810	0.0130	3.811
HiT-SRF [7]		37.794	0.9248	3.021	0.9795	0.0131	3.997
Ours		38.283	0.9319	2.645	0.9826	0.0123	3.771
Bicubic		33.102	0.8226	5.044	0.9426	0.0224	3.394
GDRRN [18]		32.441	0.8135	5.452	0.9324	0.0241	3.656
SSPSR [41]		33.753	0.8480	4.321	0.9492	0.0214	3.159
EUNet [17]		33.857	0.8467	4.448	0.9508	0.0216	3.119
AS3UNet [20]	×8	32.907	0.8179	5.638	0.9387	0.0234	3.480
MSDformer [42]		33.557	0.8340	4.643	0.9474	0.0213	3.224
ESSAformer [23]		33.852	0.8468	4.319	0.9506	0.0212	3.168
SRDNet [22]		33.836	0.8425	4.569	0.9509	0.0205	3.128
CST [19]		33.791	0.8405	4.592	0.9504	0.0217	3.143
HiT-SRF [7]		33.270	0.8251	5.381	0.9437	0.0220	3.330
Ours		33.876	0.8471	4.301	0.9506	0.0210	3.115

TABLE IV

ABLATION STUDY OF THE PROPOSED NETWORK STRUCTURES

Method	configuration	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
A	Ours w/o LRR	44.155	0.9873	2.422	0.9979	0.0070	15.702
B	A + linear LRR	44.270	0.9881	2.429	0.9985	0.0068	13.274
C	Ours w/o SpecSE	44.172	0.9877	2.571	0.9983	0.0069	14.366
D	Ours w/o SCAM	43.865	0.9874	2.484	0.9970	0.0074	15.012
E	Ours	44.371	0.9887	2.412	0.9988	0.0065	10.982

1) Effectiveness of the Proposed LRR: To evaluate the effectiveness of the proposed LRR module, we conduct ablation experiments on the LRSAM framework. As shown in Table IV, removing LRR (Method A) leads to a noticeable performance drop across all evaluation metrics compared with the full model (Method E), indicating that explicitly modeling spectral low-rank structure is beneficial for HSI SR. To further assess the role of nonlinearity, we also evaluate a linear LRR variant (Method B), in which the softmax nonlinearity is removed, and the basis vectors are combined linearly. Although the linear LRR partially recovers the performance loss of Method A, it still falls short of the proposed nonlinear LRR (Method E), which consistently achieves the best PSNR, SSIM, SAM, CC, RMSE, and ERGAS. These results demonstrate that the proposed nonlinear LRR not only exploits spectral low-rank properties but also provides additional modeling capacity beyond linear LRRs.

2) Effectiveness of the Proposed LRS2SAT: To evaluate the contribution of the proposed LRS2SAT, we remove the SpecSE module and directly concatenate the features from LRSAM and SpeAtten. As shown in Table IV, removing SpecSE (Method C) leads to a consistent and noticeable degradation across all evaluation metrics compared with the full model, confirming the effectiveness of the proposed spectral-to-spatial attention. This is because spectral cues can better guide spatial feature refinement, thereby improving the overall representational capacity of the network.

3) Effectiveness of the Proposed SCAM: To assess the effectiveness of the proposed SCAM module, we incorporate SCAM into each HFCT component and compare it with the variant without SCAM (Method D). As shown in Table IV, introducing SCAM consistently improves all evaluation metrics, demonstrating its effectiveness. We attribute these gains to SCAM's convolutional design, which enhances the modeling of high-frequency details and thus strengthens feature representation.

TABLE V

ANALYSIS OF THE SIZE OF THE LEARNED LRR

Size	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
(128, 6)	44.278	0.9875	2.444	0.9980	0.0072	14.300
(128, 8)	44.363	0.9882	2.427	0.9984	0.0070	13.128
(128, 12)	**44.371**	**0.9887**	**2.412**	**0.9988**	**0.0065**	10.982
(128, 16)	44.346	0.9876	2.436	0.9980	0.0067	**10.918**
(128, 22)	44.336	0.9874	2.453	0.9984	0.0069	11.304
(128, 30)	44.138	0.9874	2.427	0.9977	0.0073	16.238
(64, 12)	44.195	0.9870	2.429	0.9978	0.0071	13.176
(64, 16)	44.263	0.9873	2.425	0.9975	0.0070	15.026
(64, 22)	43.969	0.9870	2.430	0.9970	0.0075	15.282

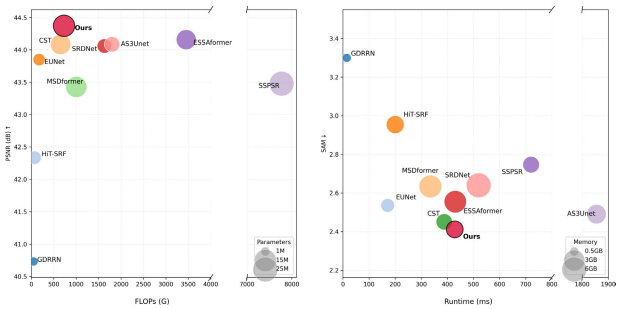
4) Analysis of the Size of the Learned LRR: The LRR is crucial for modeling low-rank characteristics, and its dimensional configuration can substantially affect SR performance. We therefore study the impact of the LRR size by varying its dimension M and the number of bases N. Here, N characterizes the intrinsic spectral dimensionality of an HSI, i.e., the number of basis spectra needed to approximate its low-rank structure. As reported in Table V, the best performance is achieved with $M = 128$ and $N = 12$. Increasing M generally improves the results, indicating that a higher feature dimension enhances representational capacity. By contrast, increasing N beyond a moderate value leads to degraded performance, suggesting that an overly large rank introduces redundancy and weakens the LRR.

5) Analysis of the Parameter \alpha: Equation (26), the scaling coefficient α is introduced to balance the contributions of the low-rank feature V_{LR} and the original feature V in the final feature fusion. To evaluate its impact, we test α under six settings, namely 0.05, 0.1, 0.2, 0.3, 0.5, and 1. As shown in Table VI, the best performance is achieved when $\alpha = 0.1$,

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TABLE VI
ABLATION STUDY OF THE PARAMETER α

α	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	EGRAS↓
0.05	44.210	0.9871	2.425	0.9984	0.0067	11.866
0.1	**44.371**	**0.9887**	**2.412**	**0.9988**	**0.0065**	10.982
0.2	44.200	0.9872	2.418	0.9981	0.0067	13.258
0.3	44.148	0.9874	2.406	0.9979	0.0068	12.928
0.5	44.333	0.9876	2.407	0.9983	0.0066	11.562
1.0	44.228	0.9873	2.432	0.9983	0.0067	**10.128**



(a) (b)

Fig. 10. Efficiency comparison of different methods on CAVE dataset (×2). (a) PSNR–FLOPs trade-off, where the bubble size denotes the number of parameters. (b) SAM–runtime trade-off, where the bubble size denotes the peak GPU memory consumption.

suggesting that an appropriate contribution of the low-rank feature provides a better trade-off between low-rank regularization and detail preservation.

G. Efficiency Comparison

To provide a more comprehensive evaluation of computational efficiency, we compare the proposed method with representative baseline methods in terms of parameter count, floating point operations (FLOPs), inference runtime, and memory consumption. For fairness, all methods are evaluated under the same experimental setting using an input size of $1 \times 31 \times 256 \times 256$ on NVIDIA RTX 3090 GPU. The parameter count is obtained by summing all learnable parameters in each model. The FLOPs are evaluated from a single forward pass with the same input size for all methods. The inference runtime is measured over 200 iterations after 50 warm-up runs, and the average runtime per iteration is reported. The memory consumption is measured as the peak GPU memory allocated during inference.

The efficiency comparison results on the CAVE dataset at a scale factor of 2 are shown in Fig. 10(a) and Fig. 10(b). Fig. 10(a) illustrates the trade-off among PSNR, FLOPs, and parameter count, while Fig. 10(b) shows the relationship among SAM, inference runtime, and memory consumption. As observed, the proposed method achieves a favorable balance between reconstruction accuracy and computational efficiency.

H. Local eRank Evaluation

This subsection investigates the rank characteristics of HSIs in the context of SR. Specifically, we aim to determine:

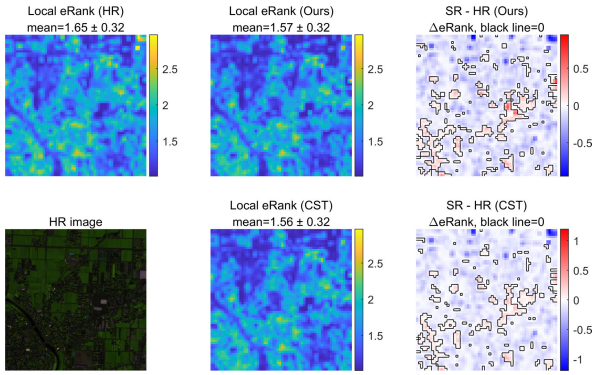


Fig. 11. Local effective-rank maps and difference maps on the test image of the Chikusei dataset.

1) whether the reconstructed images exhibit spectral low-rank structure and 2) to what extent their local spectral dimensionality agrees with the HR reference. To this end, we use the eRank to quantify local spectral dimensionality and to generate spatially resolved maps for analysis. Specifically, experiments are conducted on the test image from the Chikusei dataset at a scale factor of ×4.

Given a tensor $X \in \mathbb{R}^{H\times W\times B}$ with B spectral bands, we extract a local patch of size $w \times w \times B$, reshape it into a matrix $M \in \mathbb{R}^{N\times B}$ with $N = w^2$, and compute its singular values $\{\sigma_i\}$. We define the normalized spectral-energy distribution

$$p_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2}, \qquad H(p) = -\sum_i p_i \ln p_i$$

and the eRank, which estimates the local spectral dimensionality, as

$$r_{\text{eff}}(M) = \exp(H(p)).$$

Intuitively, $r_{\text{eff}} \approx 1$ indicates a single dominant spectral component, whereas $r_{\text{eff}} \approx 2$ suggests two comparable components within the window. We slide a window of size $w = 16$ with stride 8 over X. For each window, we subtract the per-band mean and perform ℓ_2 normalization by the Frobenius norm before computing the SVD. The window yields a scalar effective-rank value, which is assigned to the window region, and averaged over overlaps to produce a local effective-rank map $E(X) \in \mathbb{R}^{H\times W}$. To assess rank consistency, we compute the difference

$$\Delta r(x, y) = E(\text{SR})(x, y) - E(\text{HR})(x, y)$$

and visualize it as a spatial map using a diverging colormap, where blue indicates $\Delta r < 0$, white denotes $\Delta r = 0$, and red corresponds to $\Delta r > 0$. As shown in Fig. 11, both SR results exhibit a lower average eRank than the HR reference, whereas our method remains closer to HR overall. For our method, positive deviations (red) are mainly concentrated around edges and textured regions, suggesting improved preservation of heterogeneous structures and fine details while maintaining a globally low-rank spectral representation. Compared with CST [19], the deviation pattern produced by our method better aligns with the structural characteristics of the HR reference, indicating a more favorable trade-off between rank reduction and detail preservation.

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VI. Conclusion

In this article, we investigated HSI SR from the perspectives of nonlinear LRR learning and spectral–spatial interaction learning. The experimental results indicate that incorporating a nonlinear and adaptive LRR is beneficial for characterizing complex spectral correlations and improving reconstruction quality. In addition, embedding spectral priors into spatial feature learning through the proposed spectral-to-spatial aggregation mechanism provides an effective way to strengthen spectral–spatial interaction and improve spatial detail reconstruction. Based on these observations, we developed an LRS2SAT for HSI SR, in which the proposed LRR and spectral convolution attention collaboratively enhance the modeling of spectral correlations, spatial structures, and high-frequency details. Extensive experiments on multiple benchmark datasets verify the effectiveness of the proposed design and demonstrate its superiority over existing methods in both quantitative performance and visual quality. In future work, we will further investigate more efficient and expressive low-rank spectral–spatial modeling strategies by drawing inspiration from conventional low-rank methods, beyond the current bicubic degradation setting, with particular interest in their applicability to real-world HSI restoration under more complex and unknown degradations.

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